

ACCESSPATH

Minor Project 2 Submission

CSCE 547 - Windows Programming

GRADUATE SUBMISSION

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Application	AccessPath - WPF Campus Accessibility and Route Management
Submission Date	October 3, 2026
GitHub Repository	https://github.com/dswain07/AccessPath-WPF

Desktop application built with WPF, C#, SQL Server LocalDB, and ADO.NET.

1. Project Overview

AccessPath is a WPF desktop application for managing campus buildings, accessible routes, building accessibility information, and route reports. The application uses a separate data-layer Class Library and a SQL Server LocalDB database. The final interface uses a single-window navigation model so Buildings, Routes, Accessibility, Administration, and Settings are displayed within one application shell.

2. Implementation Summary

Requirement / Area	Implementation
Desktop technology	WPF with C# and XAML
Database	SQL Server LocalDB / AccessPath
Database size	7 tables - satisfies the graduate requirement of 5 or more tables
Separate data layer	AccessPath.Data Class Library project
Full CRUD	Buildings, Routes, and Building Accessibility
Search	Building search by building name and building type
Settings	Runtime Light/Dark theme switching without application restart
Administration	Route-report review and status updates
Optional security feature	Admin-only authentication using PBKDF2 password hashing

3. Application Architecture

Application flow

AccessPath.UI (WPF Views) -> AccessPath.Data (Class Library) -> Repository classes /
Microsoft.Data.SqlClient -> SQL Server LocalDB

The UI project contains the application shell and views. The data project contains models, connection management, repositories, and password-hashing utilities. The separation keeps database operations out of the WPF presentation layer and satisfies the graduate requirement for a separate Class Library data layer.

4. Required Reflection Questions

4.1 Describe a challenge you overcame in this project.

One of the main challenges was restructuring the working application from multiple WPF windows into a single-window navigation architecture without breaking the existing database operations. Initially, Routes, Accessibility, Administration, and Settings opened as separate windows. I converted those interfaces into UserControls and displayed them through a ContentControl in MainWindow. During the conversion I had to preserve the existing control names, event handlers, repository calls, and theme resources. This required careful troubleshooting of XAML/code-behind class-name mismatches, event-handler references, and control templates. The result is a cleaner single-window application while preserving the existing CRUD and database functionality.

4.2 Describe a technique you learned in this project.

A key technique I learned was using the repository pattern with ADO.NET to keep data access separate from the user interface. Each major entity has a repository that opens a SQL connection, executes parameterized commands, reads records with SqlDataReader, and maps database values into model objects. I also learned how WPF ResourceDictionary files and DynamicResource references can be used for application-wide styling and runtime Light/Dark theme switching. These techniques made the application easier to organize and modify.

4.3 What more would you want to refine in your project to make it ready to sell, ship, and deploy?

To make AccessPath production-ready, I would move the database from LocalDB to a managed production SQL Server environment and move the connection string into secure configuration rather than source code. I would add automated unit and integration tests, structured logging, stronger authentication and session management, and deployment packaging/installer support. The application could also be expanded with map-based routing, richer accessibility information, user preferences, and additional accessibility testing. Finally, I would perform usability testing with target users and validate the application against Windows accessibility guidelines before release.

4.4 Which technology did you choose (WPF, WinForms, WinUI) and why?

I chose WPF because it provides a strong combination of C# application development, XAML-based UI design, data binding, reusable styles, and flexible layout controls. WPF was particularly useful for AccessPath because the project required a desktop interface with multiple data-entry forms, DataGrids, ComboBoxes, reusable theme resources, and runtime Light/Dark theme switching. The final single-window design also benefits from WPF UserControls and ContentControl navigation. WinForms would have required more manual styling for the modern interface, while WinUI 3 would introduce a newer UI stack that was not necessary for the requirements of this project.

5. Graduate Requirements

5.1 Database requirement

The AccessPath database contains seven tables, exceeding the graduate requirement of five or more tables.

5.2 Separate Class Library data layer

The data layer is implemented as the separate AccessPath.Data Class Library project. The following steps were used to make the UI project use the separate project:

- Created the WPF UI project and a separate AccessPath.Data Class Library within the same Visual Studio solution.
- Added a Project Reference from AccessPath.UI to AccessPath.Data.
- Moved database models, DatabaseConnection, and repository classes into AccessPath.Data.
- Added the Microsoft.Data.SqlClient dependency to the data project so SQL Server access remained inside the data layer.
- Imported AccessPath.Data.Models and AccessPath.Data.Repositories into the WPF views that use them.
- Set AccessPath.UI as the startup project while keeping database operations in AccessPath.Data.
- Built and tested the complete solution to verify that the WPF UI could call the repositories through the project reference.

5.3 Light and Dark themes

The application implements separate LightTheme.xaml and DarkTheme.xaml ResourceDictionaries. The Settings view allows the user to switch themes while the application is running. ThemeManager replaces the active resource dictionary, and DynamicResource references allow existing views and controls to update without relaunching the application.

5.4 Optional Admin password feature

The optional password feature was implemented for the Administration section. Administration first presents an embedded login view. The application retrieves the user by username, verifies the password against a PBKDF2 hash stored in the Users table, and checks that the authenticated account has the Admin role before loading the Administration view. Passwords are not stored as plaintext.

6. Required Screenshot Evidence

6.1 AccessPath Start Window

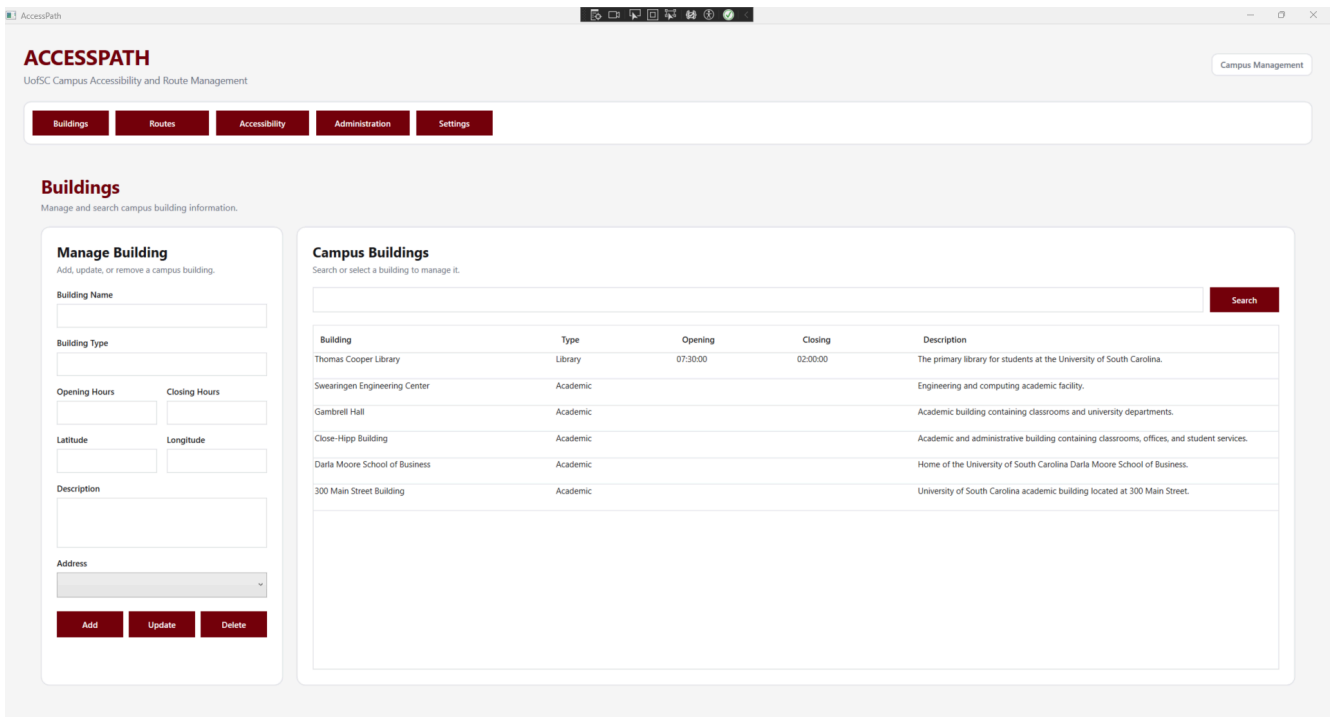


Figure 1. Starting Window

6.2 Search functionality on Building Table

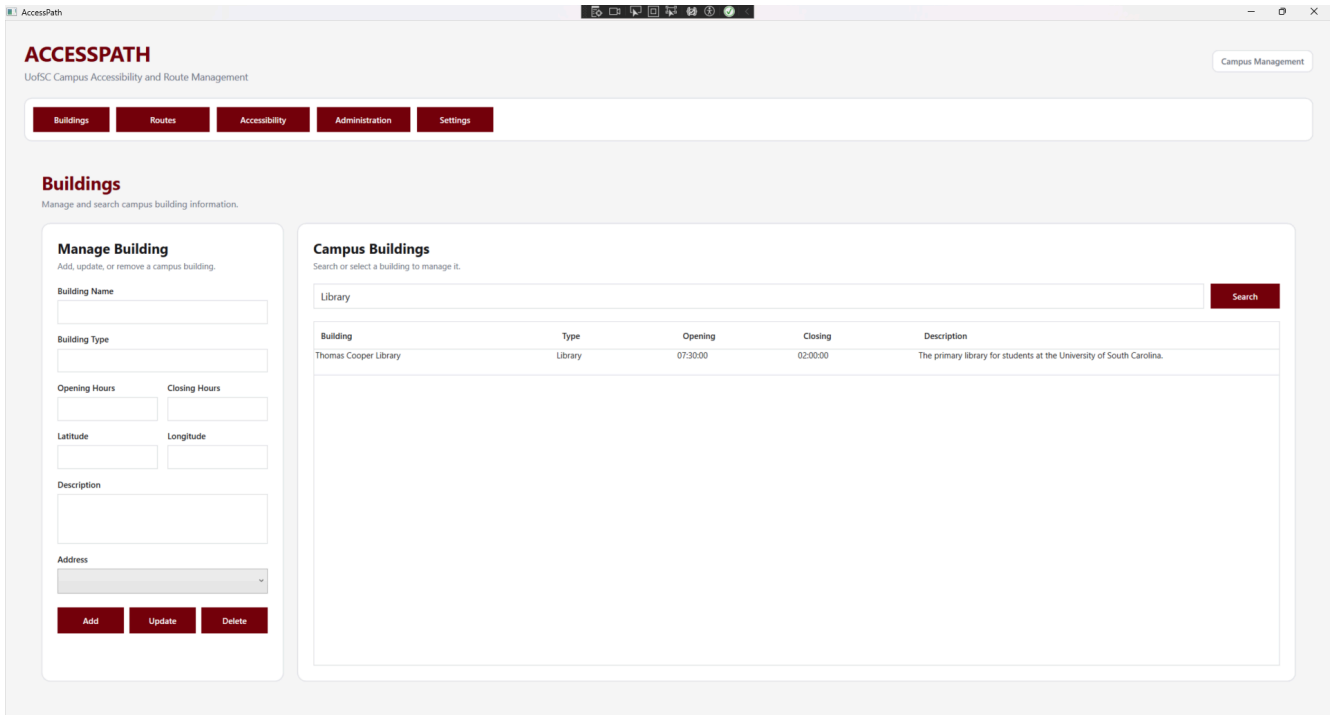


Figure 1: Search using Type of Building

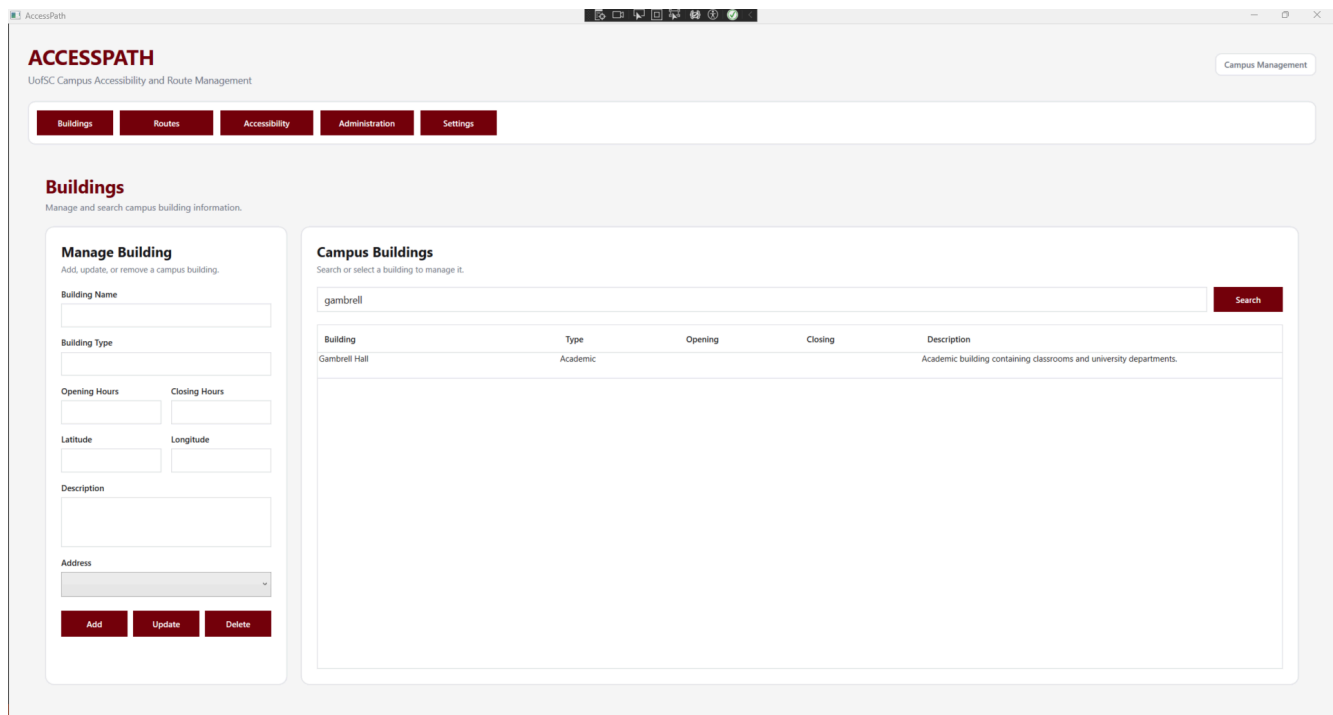


Figure 2. Search using Building Name

6.3 Create functionality on 3 Tables (Buildings, Routes, Accessibility)

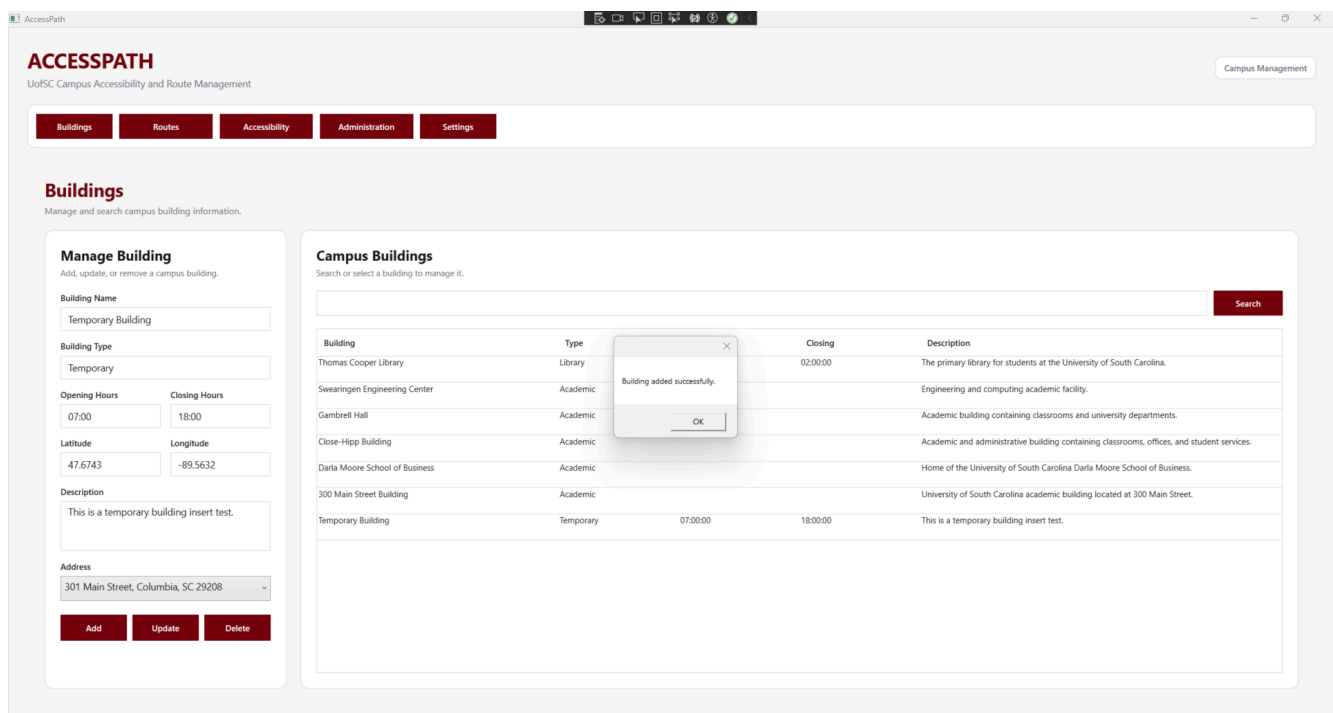


Figure 1. Create/Add on Buildings Table

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UofSC Campus Accessibility and Route Management

Campus Management

Buildings

Routes

Accessibility

Administration

Settings

Routes

Manage accessible routes between campus buildings.

Manage Route

Add, update, or remove a campus route.

Start Building

Close-Hipp Building

Destination Building

300 Main Street Building

Estimated Minutes

14

Distance

0.9

Route Status

Under Maintenance

Description

This is a temporary insert test.

Add

Update

Delete

Campus Routes

Select a route to edit or delete it.

Minutes	Distance	Status	Description	Start ID	Destination ID
12	0.600000	Active	Route from Thomas Cooper Library to Swearingen Engineering Center.	1	2
8	0.400000	Active	Route from Thomas Cooper Library to Gambrell Hall.	1	3
10	0.500000	Active	Route from Thomas Cooper Library to Close-Hipp Building.	1	4
15	0.800000	Active	Route from Thomas Cooper Library to Darla Moore School of Business.	1	5
9	0.450000	Active	Route from Gambrell Hall to Close-Hipp Building.	3	4
14	0.900000	Under Maintenance	This is a temporary insert test.	4	6

Figure 2. Create/Add on Routes Table

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UofSC Campus Accessibility and Route Management

Campus Management

Buildings

Routes

Accessibility

Administration

Settings

Accessibility

Manage accessibility features available in campus buildings.

Manage Accessibility

Add, update, or remove an accessibility feature.

Building

Thomas Cooper Library

Accessibility Feature

Temporary Feature

Description

This is a temporary insert test.

Add

Update

Delete

Building Accessibility Features

Select a feature to edit or delete it.

Feature	Description	Building ID
Accessible Entrance	Building provides an entrance designed for wheelchair accessibility.	1
Elevator	Elevator access is available for movement between floors.	1
Accessible Restroom	Accessible restroom facilities are available within the building.	1
Accessible Entrance	Building provides an accessible entrance for users with mobility needs.	2
Elevator	Elevator access is available for movement between floors.	2
Wheelchair Ramp	A wheelchair-accessible ramp provides an alternative to stairs.	2
Accessible Entrance	Building provides an entrance designed for wheelchair accessibility.	3
Elevator	Elevator access is available for movement between floors.	3
Accessible Entrance	Building provides an accessible entrance for users with mobility needs.	4
Elevator	Elevator access is available for movement between floors.	4
Accessible Restroom	Accessible restroom facilities are available within the building.	4
Accessible Entrance	Building provides an entrance designed for wheelchair accessibility.	5
Automatic Door	An automatic door assists users entering the building.	5
Elevator	Elevator access is available for movement between floors.	5

Figure 3. Create/Add on Accessibility Table

6. Required Screenshot Evidence (continued)

6.4 Update functionality 3 Tables (Buildings, Routes, Accessibility)

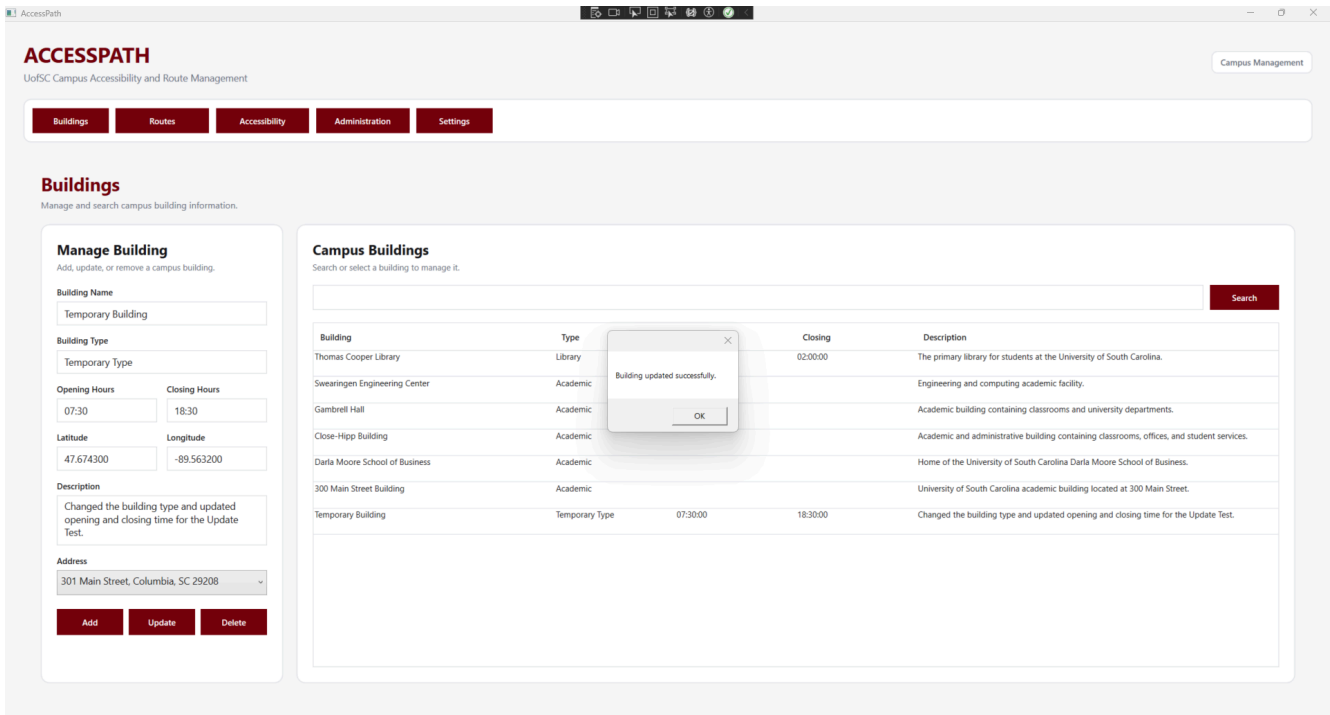


Figure 1. Update on Buildings Table

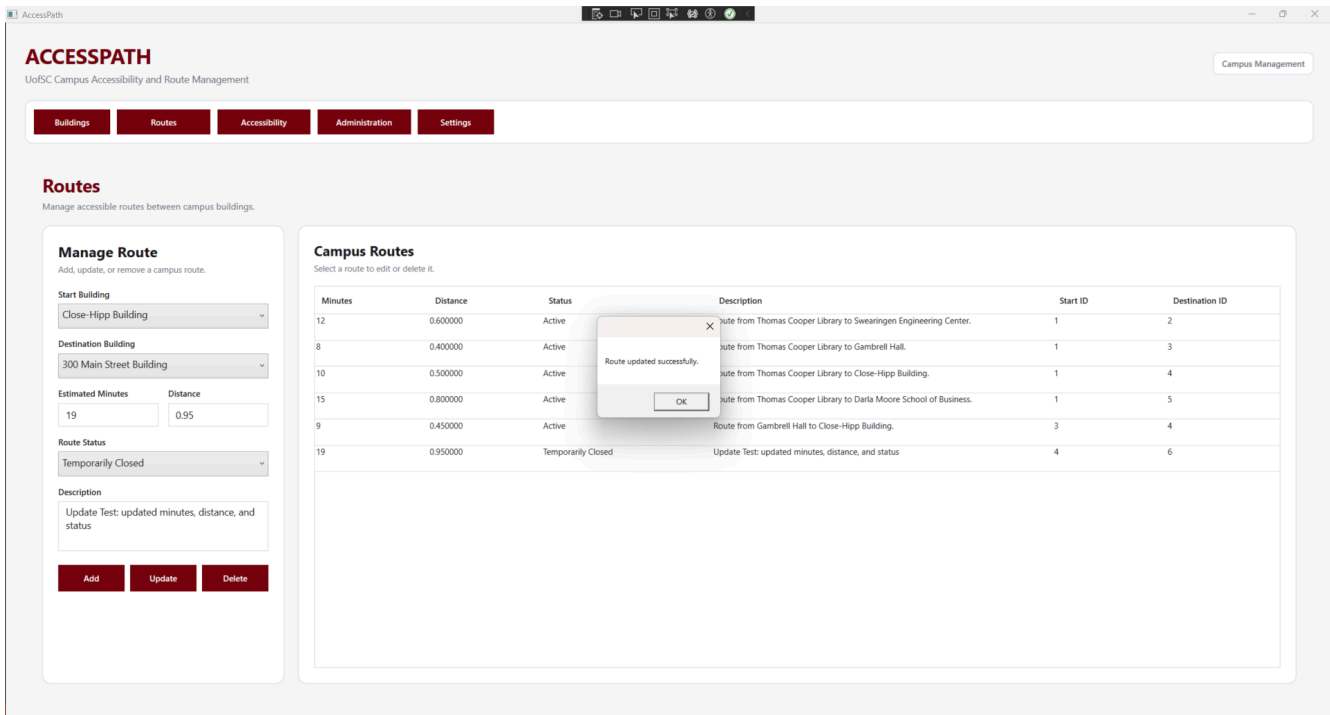


Figure 2. Update on Routes Table

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UofSC Campus Accessibility and Route Management

Campus Management

Buildings

Routes

Accessibility

Administration

Settings

Accessibility

Manage accessibility features available in campus buildings.

Manage Accessibility

Add, update, or remove an accessibility feature.

Building

Thomas Cooper Library

Accessibility Feature

Updating Feature

Description

Update Test

Add

Update

Delete

Building Accessibility Features

Select a feature to edit or delete it.

Feature	Description	Building ID
Wheelchair Ramp	A wheelchair-accessible ramp provides an alternative to stairs.	2
Accessible Entrance		3
Elevator		3
Accessible Entrance		4
Elevator		4
Accessible Restroom	Accessible restroom facilities are available within the building.	4
Accessible Entrance	Building provides an entrance designed for wheelchair accessibility.	5
Automatic Door	An automatic door assists users entering the building.	5
Elevator	Elevator access is available for movement between floors.	5
Accessible Restroom	Accessible restroom facilities are available within the building.	5
Accessible Entrance	Building provides an accessible entrance for users with mobility needs.	6
Elevator	Elevator access is available for movement between floors.	6
Updating Feature	Update Test	1

Accessibility feature updated successfully.

OK

Figure 3. Update on Accessibility Table

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6.5 Delete functionality 3 Tables (Buildings, Routes, Accessibility)

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UofSC Campus Accessibility and Route Management

Campus Management

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Accessibility

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Accessibility

Manage accessibility features available in campus buildings.

Manage Accessibility

Add, update, or remove an accessibility feature.

Building

Thomas Cooper Library

Accessibility Feature

Updating Feature

Description

Update Test

Add

Update

Delete

Building Accessibility Features

Select a feature to edit or delete it.

Feature	Description	Building ID
Wheelchair Ramp	A wheelchair-accessible ramp provides an alternative to stairs.	2
Accessible Entrance	Building provides an entrance designed for wheelchair accessibility.	3
Elevator	Elevator access is available for movement between floors.	3
Accessible Entrance	Building provides an accessible entrance for users with mobility needs.	4
Elevator	Elevator access is available for movement between floors.	4
Accessible Restroom	Accessible restroom facilities are available within the building.	4
Accessible Entrance	Building provides an entrance designed for wheelchair accessibility.	5
Automatic Door	An automatic door assists users entering the building.	5
Elevator	Elevator access is available for movement between floors.	5
Accessible Restroom	Accessible restroom facilities are available within the building.	5
Accessible Entrance	Building provides an accessible entrance for users with mobility needs.	6
Elevator	Elevator access is available for movement between floors.	6
Updating Feature	Update Test	1

Confirm Delete

Are you sure you want to delete Updating Feature?

Yes

No

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Manage accessibility features available in campus buildings.

Manage Accessibility

Add, update, or remove an accessibility feature.

Building

Thomas Cooper Library

Accessibility Feature

Updating Feature

Description

Update Test

Add

Update

Delete

Building Accessibility Features

Select a feature to edit or delete it.

Feature	Description	Building ID
Elevator	Elevator access is available for movement between floors.	2
Wheelchair Ramp	A wheelchair-accessible ramp provides an alternative to stairs.	2
Accessible Entrance	Building provides an entrance designed for wheelchair accessibility.	3
Elevator	Elevator access is available for movement between floors.	3
Accessible Entrance	Building provides an accessible entrance for users with mobility needs.	4
Elevator	Elevator access is available for movement between floors.	4
Accessible Restroom	Accessible restroom facilities are available within the building.	4
Accessible Entrance	Building provides an entrance designed for wheelchair accessibility.	5
Automatic Door	An automatic door assists users entering the building.	5
Elevator	Elevator access is available for movement between floors.	5
Accessible Restroom	Accessible restroom facilities are available within the building.	5
Accessible Entrance	Building provides an accessible entrance for users with mobility needs.	6
Elevator	Elevator access is available for movement between floors.	6

Accessibility feature deleted successfully.

OK

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Buildings

Routes

Accessibility

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Settings

Buildings

Manage and search campus building information.

Manage Building

Add, update, or remove a campus building.

Building Name

Temporary Building

Building Type

Temporary Type

Opening Hours

07:30

Closing Hours

18:30

Latitude

47.674300

Longitude

-89.563200

Description

Changed the building type and updated opening and closing time for the Update Test.

Address

301 Main Street, Columbia, SC 29208

Add

Update

Delete

Campus Buildings

Search or select a building to manage it.

Search

Building	Closing	Description
Thomas Cooper Library	02:00:00	The primary library for students at the University of South Carolina.
Swearingen Engineering Center		Engineering and computing academic facility.
Gambrell Hall		Academic building containing classrooms and university departments.
Close-Hipp Building		Academic and administrative building containing classrooms, offices, and student services.
Darla Moore School of Business		Home of the University of South Carolina Darla Moore School of Business.
300 Main Street Building		University of South Carolina academic building located at 300 Main Street.
Temporary Building	Temporary type	07:30:0018:30:00Changed the building type and updated opening and closing time for the Update Test.

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UofSC Campus Accessibility and Route Management

Campus Management

Buildings

Routes

Accessibility

Administration

Settings

Buildings

Manage and search campus building information.

Manage Building

Add, update, or remove a campus building.

Building Name

Building Type

Opening Hours

Closing Hours

Latitude

Longitude

Description

Address

Add

Update

Delete

Campus Buildings

Search or select a building to manage it.

Search

Building	Type	Closing	Description
Thomas Cooper Library	Library	02:00:00	The primary library for students at the University of South Carolina.
Swearingen Engineering Center	Academic		Engineering and computing academic facility.
Gambrell Hall	Academic		Academic building containing classrooms and university departments.
Close-Hipp Building	Academic		Academic and administrative building containing classrooms, offices, and student services.
Darla Moore School of Business	Academic		Home of the University of South Carolina Darla Moore School of Business.
300 Main Street Building	Academic		University of South Carolina academic building located at 300 Main Street.

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Settings

Routes

Manage accessible routes between campus buildings.

Manage Route

Add, update, or remove a campus route.

Start Building

Close-Hipp Building

Destination Building

300 Main Street Building

Estimated Minutes

19

Distance

0.950000

Route Status

Temporarily Closed

Description

Update Test: updated minutes, distance, and status

Add

Update

Delete

Campus Routes

Select a route to edit or delete it.

Minutes	Distance	Status	Description	Start ID	Destination ID
12	0.600000	Active	Route from Thomas Cooper Library to Swearingen Engineering Center.	1	2
8	0.400000	Active	Route from Thomas Cooper Library to Gambrell Hall.	1	3
10	0.500000	Active	Route from Thomas Cooper Library to Close-Hipp Building.	1	4
15	0.800000	Active	Route from Thomas Cooper Library to Darla Moore School of Business.	1	5
9	0.450000	Active	Route from Gambrell Hall to Close-Hipp Building.	3	4
19	0.950000	Temporarily Closed	Update test: updated minutes, distance, and status	4	6

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Routes

Accessibility

Administration

Settings

Routes

Manage accessible routes between campus buildings.

Manage Route

Add, update, or remove a campus route.

Start Building

Close-Hipp Building

Destination Building

300 Main Street Building

Estimated Minutes

19

Distance

0.950000

Route Status

Temporarily Closed

Description

Update Test: updated minutes, distance, and status

Add

Update

Delete

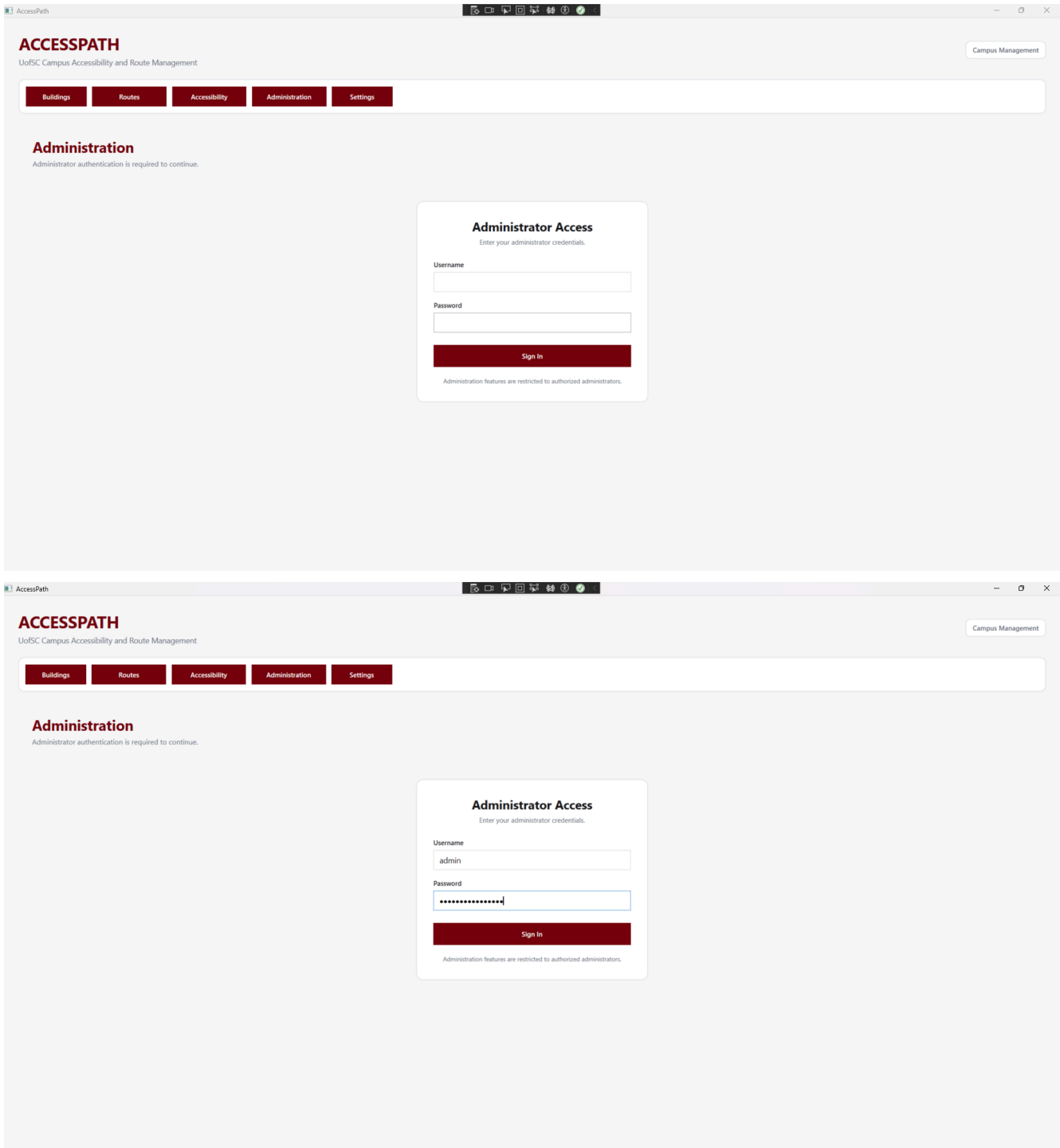
Campus Routes

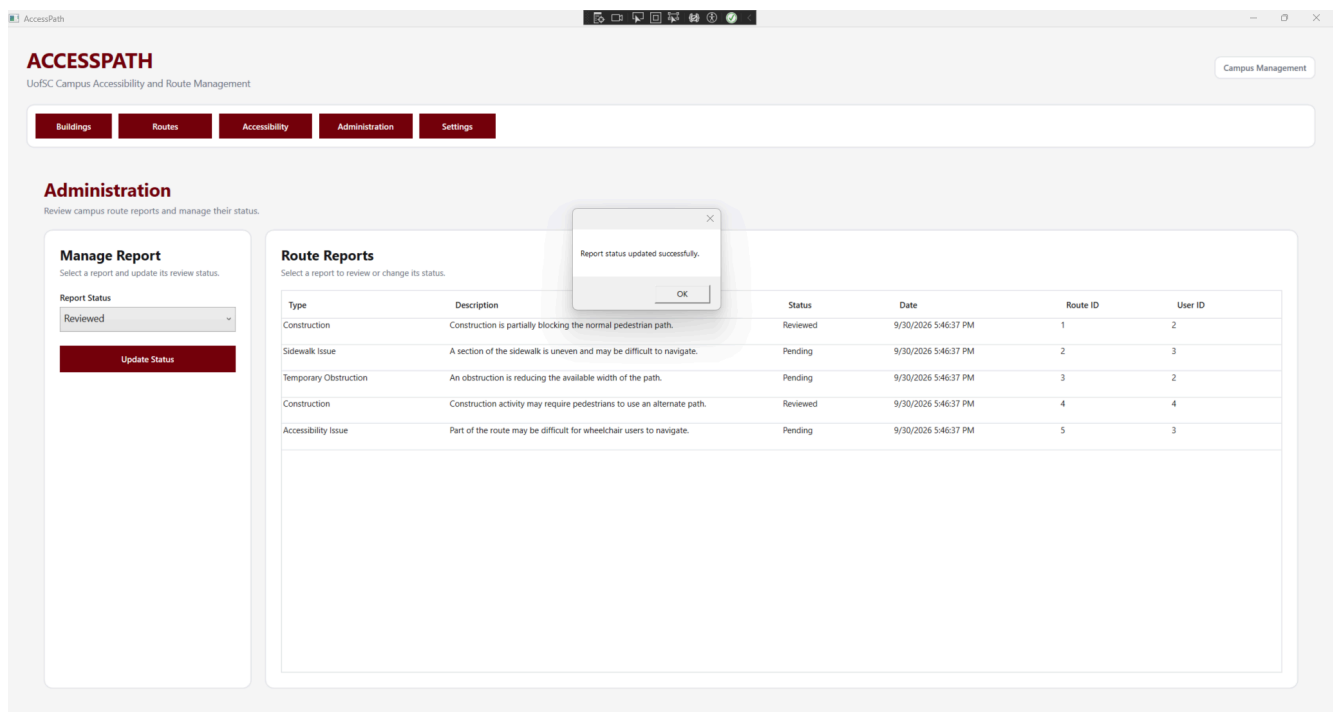
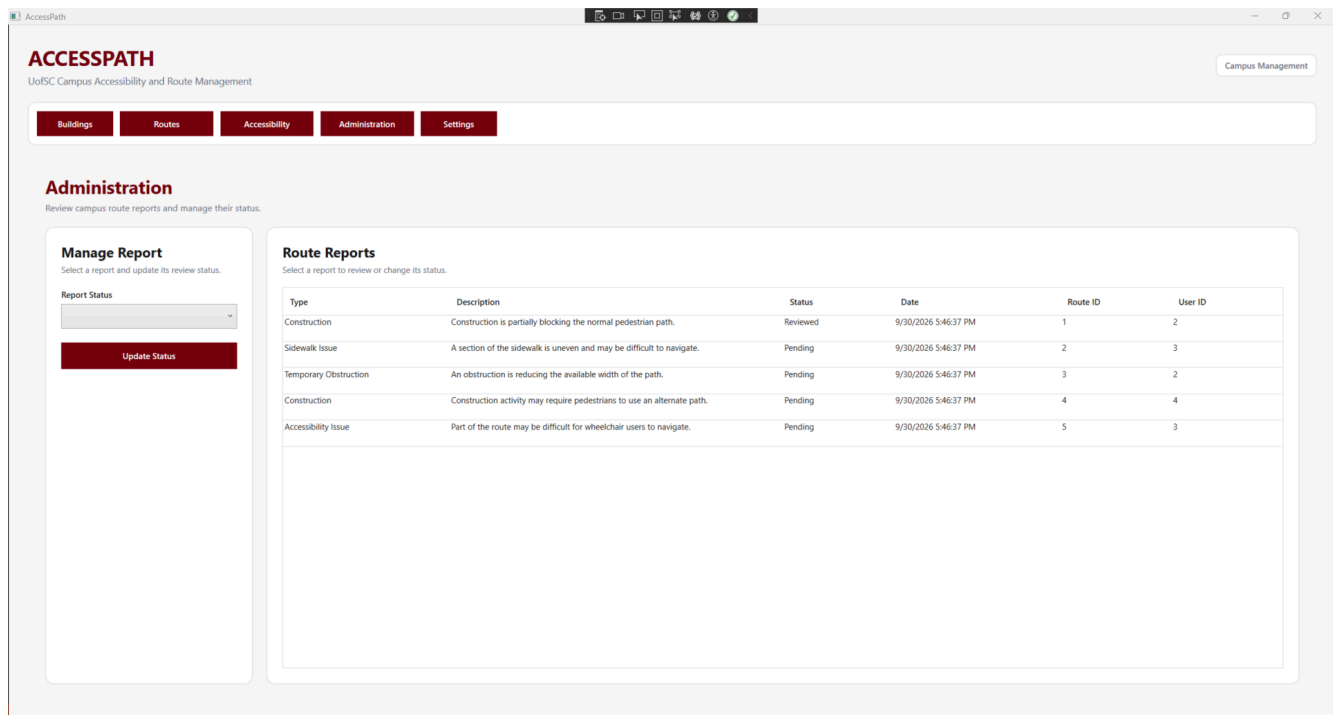
Select a route to edit or delete it.

Minutes	Distance	Status	Description	Start ID	Destination ID
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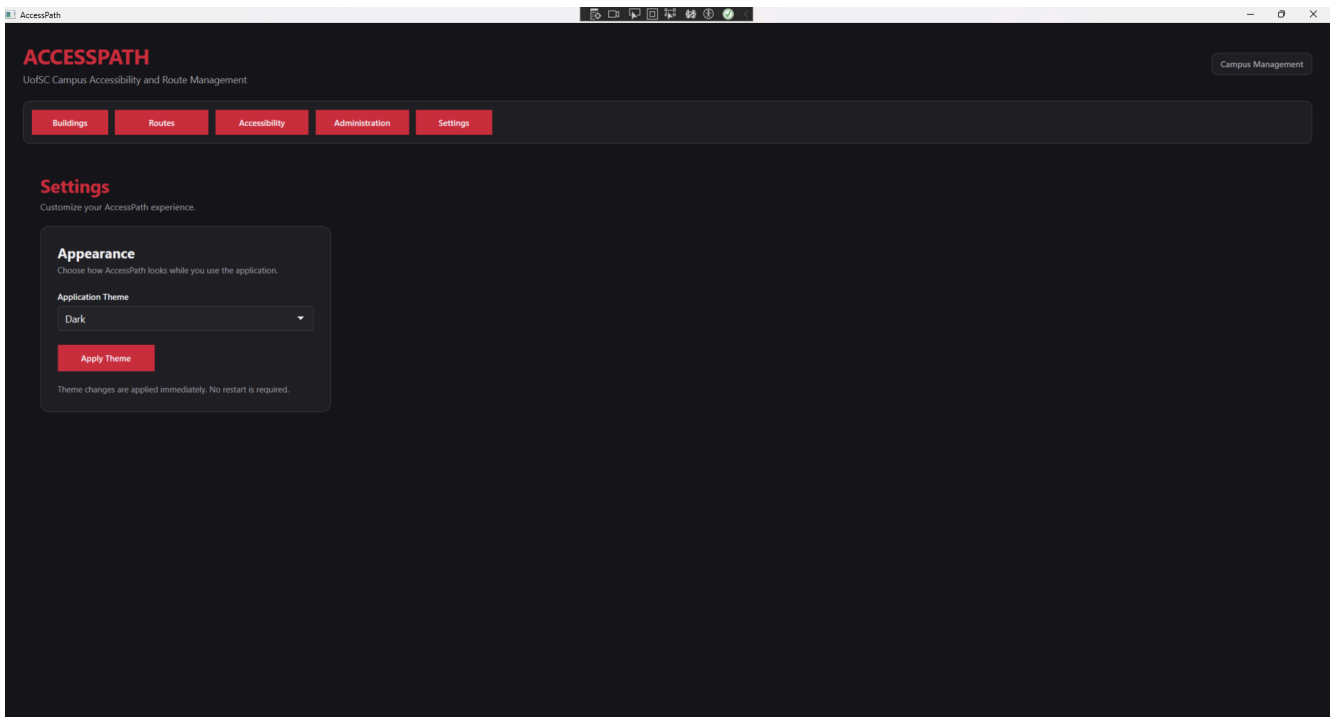
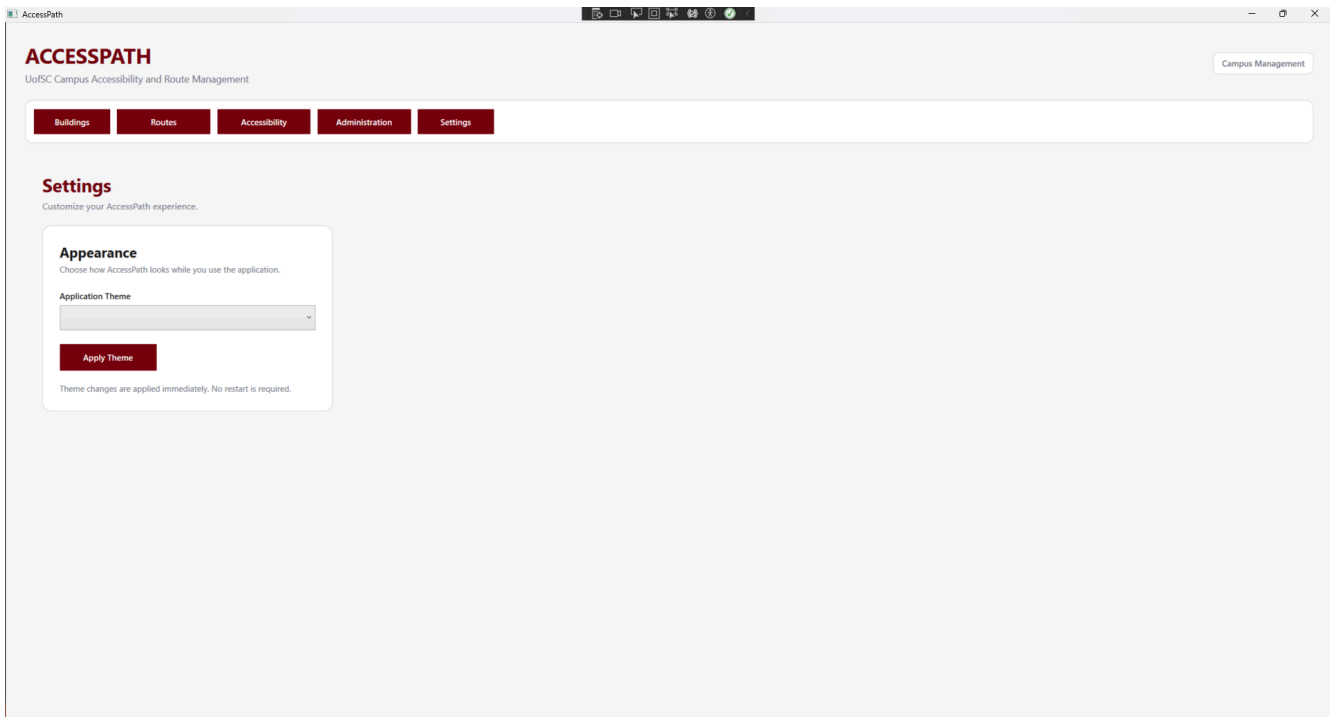
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6.6 Administration Page, Sign-in, Update Report Table





6.7 Settings Page, Dark Theme



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Buildings

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Add, update, or remove a campus building.

Building Name

Building Type

Opening Hours

Closing Hours

Latitude

Longitude

Description

Address

Add

Update

Delete

Campus Buildings

Search or select a building to manage it.

Search

Building	Type	Opening	Closing	Description
Thomas Cooper Library	Library	07:30:00	02:00:00	The primary library for students at the University of South Carolina.
Swearingen Engineering Center	Academic			Engineering and computing academic facility.
Gambrell Hall	Academic			Academic building containing classrooms and university departments.
Close-Hipp Building	Academic			Academic and administrative building containing classrooms, offices, and student services.
Darla Moore School of Business	Academic			Home of the University of South Carolina Darla Moore School of Business.
300 Main Street Building	Academic			University of South Carolina academic building located at 300 Main Street.

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